Multimodal Emotion Recognition using EEG, ECG, and HRV Signals on the DREAMER Dataset

1. Introduction

Emotion recognition from physiological signals has become an important research area in affective computing, human-computer interaction, mental health monitoring, and stress analysis. Physiological signals provide objective measurements of emotional responses and are less affected by intentional behavioral changes compared to facial expressions or speech.

In this project, a multimodal deep learning framework was developed for emotion recognition using the DREAMER dataset. The proposed system integrates three physiological modalities:

- Electroencephalogram (EEG)
- Electrocardiogram (ECG)
- Heart Rate Variability (HRV)

A custom multimodal Convolutional Neural Network (CNN) architecture was designed to jointly learn discriminative emotional representations from all modalities. The model was trained and evaluated for three-class valence classification.

The final framework achieved strong classification performance with:

- Accuracy: 86.43%
- Precision: 86.42%
- Recall: 86.43%
- F1-Score: 86.41%

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2. DREAMER Dataset Overview

The DREAMER dataset is a multimodal affective computing database designed for emotion recognition using physiological signals.

The dataset contains recordings from 23 participants who watched 18 emotional video stimuli designed to induce different emotional states.

2.1 Physiological Signals

EEG (Electroencephalogram)

EEG signals were acquired using the Emotiv EPOC headset with 14 channels:

- AF3
- F7
- F3
- FC5
- T7
- P7
- O1
- O2
- P8
- T8
- FC6
- F4
- F8
- AF4

Sampling frequency:

- 128 Hz

ECG (Electrocardiogram)

ECG signals were recorded using a Shimmer sensor system.

- 2 ECG channels
- Sampling frequency: 256 Hz

Subjective Emotional Scores

After each video stimulus, participants rated their emotional states using the Self-Assessment Manikin (SAM) scale.

Three affective dimensions were recorded:

- Valence
- Arousal
- Dominance

Each score ranged from 1 to 5.

3. DREAMER Data Structure

The original DREAMER dataset is distributed as a MATLAB (.mat) file.

The data hierarchy contains:

Root Structure

- DREAMER
    - Data
    - Age
    - Gender

Subject-Level Data

Each subject contains:

EEG Data

For every trial:

- stimuli → active EEG recording during emotional video playback
- baseline → resting baseline EEG recording

EEG signal shape:

- (Samples, 14)

ECG Data

For every trial:

- stimuli → active ECG recording
- baseline → resting ECG recording

ECG signal shape:

- (Samples, 2)

Emotional Labels

For each of the 18 trials:

- ScoreValence
- ScoreArousal
- ScoreDominance

4. Data Loading Pipeline

A custom DREAMER loading pipeline was implemented in Python to:

- Read the MATLAB structure
- Parse nested dictionaries
- Extract EEG and ECG recordings
- Extract emotional labels
- Organize subject-wise trial data

The loader function generated a clean dictionary-based structure:

```
{
    "subjects": [
        {
            "eeg_stimuli": [...],
            "eeg_baseline": [...],
            "ecg_stimuli": [...],
            "ecg_baseline": [...],
            "valence": [...],
            "arousal": [...],
            "dominance": [...]
        }
    ]
}
```

The preprocessing framework processed all 23 subjects and 18 emotional trials.

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5. Signal Processing and Preprocessing

Proper signal preprocessing is critical for physiological emotion recognition systems because EEG and ECG recordings contain noise, motion artifacts, and baseline drift.

The following preprocessing pipeline was implemented.

5.1 Bandpass Filtering

EEG Filtering

A Butterworth bandpass filter was applied:

- Low cutoff: 1 Hz
- High cutoff: 45 Hz

- Filter order: 4

Purpose:

- Remove DC drift
- Remove high-frequency noise
- Preserve emotion-related EEG frequency bands

ECG Filtering

ECG signals were filtered using:

- Low cutoff: 0.5 Hz
- High cutoff: 40 Hz

Purpose:

- Remove motion artifacts
- Remove baseline wander
- Preserve cardiac waveform information

5.2 Z-Score Normalization

Each signal channel was normalized independently using z-score normalization:

[X_{norm} =]

This normalization:

- Stabilizes neural network training
- Reduces inter-subject variability
- Prevents scale imbalance between modalities

5.3 Sliding Window Segmentation

Signals were segmented into fixed temporal windows.

Window Configuration

- Window duration: 10 seconds
- Overlap ratio: 50%

EEG Window Size

Since EEG was sampled at 128 Hz:

- 1280 samples per window

ECG / HRV Window Size

Since ECG was sampled at 256 Hz:

- 2560 samples per window

Overlapping windows increased dataset size and improved temporal learning.

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6. HRV Signal Extraction

In addition to raw ECG signals, a continuous Heart Rate Variability (HRV) signal was generated.

HRV is strongly related to autonomic nervous system activity and emotional regulation.

6.1 HRV Extraction Steps

Step 1: R-Peak Detection

R-peaks were detected using the SciPy peak detection algorithm.

Step 2: RR Interval Computation

The intervals between consecutive R-peaks were calculated:

[RR_i = t_{i+1} - t_i]

Step 3: Continuous HRV Interpolation

RR intervals were interpolated into a continuous waveform using linear interpolation.

The final HRV signal was used as a third physiological modality.

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7. Label Generation

The task focused on three-class valence classification.

The original 1–5 valence scores were converted into three emotional categories.

Label Mapping Strategy

Original Score	Class
1–2	Low

Original Score	Class
3	Medium
4–5	High

This transformation simplified the classification problem while preserving emotional semantics.

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8. Dataset Preparation

After preprocessing and windowing, the dataset consisted of:

- EEG windows
- ECG windows
- HRV windows
- Emotional labels

Each training sample contained:

- EEG tensor
- ECG tensor
- HRV tensor
- Valence label

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9. Proposed Multimodal CNN Architecture

A custom multimodal deep learning architecture was designed.

The model uses three parallel CNN branches:

1. EEG branch
2. ECG branch
3. HRV branch

Each branch independently extracts modality-specific temporal features.

The extracted features are then fused for final emotion classification.

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10. EEG Branch

The EEG stream processes 14-channel temporal EEG windows.

Architecture

Convolution Block 1

- Conv1D: 14 → 32
- Kernel size: 7
- Batch Normalization
- ReLU
- MaxPooling

Convolution Block 2

- Conv1D: 32 → 64
- Kernel size: 5
- Batch Normalization
- ReLU
- MaxPooling

Convolution Block 3

- Conv1D: 64 → 128
- Kernel size: 3
- Batch Normalization
- ReLU
- Adaptive Average Pooling

The EEG branch learns:

- Temporal EEG dynamics
- Frequency-related patterns
- Cross-channel emotional activity

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11. ECG Branch

The ECG stream processes cardiac electrical activity.

Architecture

Convolution Block 1

- Conv1D: 1 → 32
- Kernel size: 15

Convolution Block 2

- Conv1D: 32 → 64
- Kernel size: 9

Convolution Block 3

- Conv1D: 64 → 128
- Kernel size: 5

The ECG branch captures:

- Heartbeat morphology
- Rhythm changes
- Emotion-related cardiac patterns

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12. HRV Branch

The HRV branch processes autonomic variability information.

Architecture

Convolution Block 1

- Conv1D: 1 → 16
- Kernel size: 11

Convolution Block 2

- Conv1D: 16 → 32
- Kernel size: 7

Convolution Block 3

- Conv1D: 32 → 64
- Kernel size: 5

This branch learns:

- Heart rhythm variability

- Emotional stress patterns
- Autonomic nervous system responses

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13. Feature Fusion and Classification

The outputs of all branches were concatenated.

Fusion Feature Size

- EEG: 128 features
- ECG: 128 features
- HRV: 64 features

Total fusion vector:

- 320 features

Classifier Layers

Fully Connected Layer 1

- 320 → 256
- ReLU
- Dropout: 0.4

Fully Connected Layer 2

- 256 → 128
- ReLU
- Dropout: 0.3

Output Layer

- 128 → 3 classes

The final layer outputs probabilities for:

- Low Valence
- Medium Valence
- High Valence

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14. Training Strategy

Dataset Split

The dataset was divided into:

Split	Percentage
Training	70%
Validation	15%
Testing	15%

Stratified splitting was used to preserve class balance.

Hyperparameters

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Scheduler	ReduceLROnPlateau
Batch Size	16
Epochs	100
Loss Function	CrossEntropyLoss

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15. Evaluation Metrics

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Classification Report

Weighted averaging was used for multi-class evaluation.

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16. Experimental Results

The proposed multimodal CNN achieved strong performance for three-class valence recognition.

Overall Performance

Metric	Score
Accuracy	0.8643
Precision	0.8642
Recall	0.8643
F1-Score	0.8641

Detailed Classification Report

Class	Precision	Recall	F1-Score	Support
Low	0.8729	0.8595	0.8661	975
Medium	0.8320	0.8150	0.8234	492
High	0.8713	0.8926	0.8818	1024

Overall Statistics

Metric	Value
Accuracy	0.8643
Macro Avg F1	0.8571
Weighted Avg F1	0.8641

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17. Discussion

The obtained results demonstrate that multimodal physiological fusion significantly improves emotion recognition performance.

Several important observations were identified:

EEG Contribution

EEG signals provided strong discriminative neural representations related to emotional processing.

ECG and HRV Complementarity

Cardiac signals contributed complementary autonomic information that enhanced emotional classification.

Feature Fusion Advantage

The multimodal fusion strategy enabled the model to combine:

- Neural activity
- Cardiac dynamics
- Autonomic variability

This produced more robust emotional representations than single-modality systems.

Strong High-Valence Recognition

The model achieved the highest F1-score for the High class, indicating strong discrimination of positive emotional states.

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18. Conclusion

This project presented a multimodal deep learning framework for emotion recognition using EEG, ECG, and HRV signals from the DREAMER dataset.

A dedicated preprocessing pipeline was implemented including:

- Signal filtering
- HRV extraction
- Window segmentation
- Signal normalization

A three-stream CNN architecture was developed to learn modality-specific features and fuse them for emotion classification.

The proposed framework achieved:

- 86.43% classification accuracy
- 86.41% weighted F1-score

The results demonstrate the effectiveness of multimodal physiological learning for affective computing applications.

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19. Future Work

Several directions can further improve the framework:

- Subject-independent cross-validation
- Attention-based fusion mechanisms
- Transformer architectures
- Temporal sequence modeling using LSTM or GRU layers

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20. Technologies and Libraries

The project was implemented using:

- Python
- PyTorch
- NumPy
- SciPy
- Scikit-learn
- Matplotlib
- Seaborn

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21. Summary

This work demonstrates that combining EEG, ECG, and HRV signals within a multimodal deep learning framework can achieve high-performance emotion recognition.

The proposed CNN architecture successfully learned complementary physiological representations and achieved strong classification performance on the DREAMER dataset.

The framework provides a strong foundation for future affective computing and physiological AI research.